Exercise 2

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① List all distinct departments in the Students table

```
SELECT Distinct department
FROM Students;
```

Department
IT
HR
Finance

* distinct removes duplicates from the table

② List the average age of Students per department

```
SELECT department, AVG(age) AS avg-age
FROM students
GROUP BY department;
```

Department	Age
IT	20.5
HR	22
Finance	11,5

* with the avg(age) gives the mean of all data on the table

③ Show department with more than 1 student

```
SELECT department, Count(*) AS Student-count
FROM students
GROUP BY department
HAVING Count(*) > 1;
```

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Department	Student-count
IT	2
HR	2

*Finance have only one student, that is why we dropped it.

(4) Get all students whose age is between 21 and 23

```
SELECT student-id, name, age, department
FROM Students
WHERE age Between 21 and 23;
```

student-id	Name	age	Department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eva	22	HR

*we where looking for students whose age was between 21 and 23

(5) List all the students in the IT or HR department who are older than 21

```
SELECT student-id, name, age, department
FROM students
WHERE department IN IT, HR
AND age > 21;
```

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student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

*Charlie is restricted as his age is not greater than 21

⑥ Show total credits per department, only for department worth more than 5 total credits.

```
SELECT department, Sum(credits) AS total_credits
FROM Courses
GROUP BY department
HAVING Sum(credits) >5;
```

Department	total_Credits
IT	11

⑦ List all courses that do not have 4 Credits

```
SELECT Course_id, Course_name, department, Credits
FROM Courses
WHERE Credits <>4;
```

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

* the function <> excludes rows with Credits of equals 4 and above

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(8) Show the top 3 courses by credits in descending order

SELECT Course_id, course_name, credits
FROM Courses
Order BY Credits DESC
Limit 3;

Course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

* Sorting descending then limit 3

(9) Get the maximum, minimum, and average grade across all enrollments

SELECT Max(grade) As max_grade
       Min(grade) As min_grade
       AVG(grade) As Avg_grade
FROM enrollments;

Max_grade	Min_grade	avg_grade
90	78	84,6

*

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(10) Count how many enrollments exists per course

```
SELECT course_id, count(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

*every course has exactly one enrollment in this dataset

(11) Find total salary and total bonus per department

```
SELECT department,
       sum(salary) AS total_salary,
       sum(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7500
Finance	70 000	6000

*It sum all the rows per department with their bonuses

(12) Show departments where average salary is above 55 000

```
SELECT department, AVG(salary) AS avg_salary
FROM Salaries
GROUP BY department
HAVING AVG(Salary) > 55 000;
```

Department	avg_Salary
IT	61 000
Finance	70 000

(13) List employees whose salary plus bonus is greater than 60 000

```
SELECT employee_id, name, salary, bonus, salary +
        bonus AS total_compensation
FROM Salaries
Where salary + bonus > 60000
```

employee_id	name	salary	bonus	total compensation
1	Tom	60 000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

* Salaries and bonuses combine they are greater than 60000

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(14) Show total and average budget per department. Only include departments with average budget above 70 000

```
SELECT department,
       SUM(budget) AS total_budget
       AVG(budget) AS avg_budget
FROM projects
GROUP BY department
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

* we divide total budget of IT as its more 2 and keep finance as it is 1

(15) List all projects with budgets between 50 000 and 120 000, excluding the Marketing department

```
SELECT project_id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50 000 and 120 000
  AND department <> Marketing;
```

Project_id	Project_name	department	budget
1	AI App	IT	120 000
2	Payroll system	Finance	80 000
5	HR portal	HR	50 000